

NEDRE SOPPERO, Sweden

WMO: 020540

Lat: 68.0440N

Lon: 21.7550E

Elev: 1181

StdP: 14.08

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-26.5	-37.7	0.7	-30.9	-32.6	0.9	-26.1	19.0	24.2	17.6	24.3	1.1	230	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	73.2	54.4	69.4	53.9	66.2	52.8	59.6	64.9	57.8	63.3	56.2	61.7	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	75.2	60.1	56.0	69.6	58.5	54.0	64.9	57.3	26.7	65.0	25.5	63.5	24.4	62.0	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.8	13.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-36.6	62.2	5.1	2.2	-40.3	63.8	-43.3	65.1	-46.1	66.3	-49.9	68.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.9	5.5	6.1	15.7	27.9	40.2	51.4	56.5	52.5	43.5	29.9	17.1	11.1		
	DBStd	20.79	15.46	15.29	11.72	8.66	7.05	6.75	5.02	5.69	6.21	9.35	12.22	15.26		
	HDD50	7788	1380	1229	1063	664	315	63	7	40	212	623	986	1206		
	HDD65	12808	1845	1649	1528	1114	767	411	265	388	646	1088	1436	1671		
	CDD50	457	0	0	0	0	12	103	210	117	15	0	0	0		
	CDD65	5	0	0	0	0	0	2	3	0	0	0	0	0		
	CDH74	65	0	0	0	0	9	25	25	6	0	0	0	0		
	CDH80	4	0	0	0	0	0	4	0	0	0	0	0	0		
	WSAvg	6.0	4.8	5.4	6.7	6.9	7.3	7.0	6.5	5.4	6.4	5.8	4.8	5.2		
	PrecAvg	19.6	1.1	1.0	0.9	0.9	1.5	2.4	3.2	2.4	1.7	1.7	1.4	1.3		
(15) Precipitation	PrecMax	24.6	2.0	2.4	2.8	2.2	3.4	4.6	6.3	4.4	3.7	3.5	2.9	3.7		
	PrecMin	13.4	0.2	0.2	0.2	0.1	0.4	0.6	0.9	1.1	0.5	0.7	0.4	0.2		
	PrecStd	3.2	0.4	0.6	0.6	0.6	0.8	1.0	1.3	0.9	0.9	0.8	0.7	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	38.9	43.9	51.7	74.9	76.9	77.2	74.9	61.4	51.3	42.2	40.1		
		MCWB	32.5	33.3	36.2	40.0	51.2	56.8	57.6	55.4	49.3	42.8	37.8	34.3		
	2%	DB	33.7	34.9	39.2	46.7	64.7	71.9	73.1	70.4	58.5	48.1	37.6	37.0		
		MCWB	30.1	30.8	32.5	36.3	46.1	54.1	54.9	54.2	49.7	42.5	34.9	32.6		
	5%	DB	29.7	30.4	35.9	43.7	57.7	67.4	69.3	67.1	55.9	45.2	34.4	32.9		
		MCWB	27.7	27.8	31.1	34.6	43.3	52.1	54.7	52.9	48.9	41.3	32.9	30.6		
	10%	DB	25.8	27.2	33.0	40.8	52.3	63.2	66.2	63.7	53.6	41.9	32.6	30.7		
		MCWB	24.5	25.5	29.1	33.0	41.1	50.4	53.9	52.2	48.4	39.3	31.5	28.8		
	0.4%	WB	33.2	34.0	36.5	41.4	51.9	59.7	62.4	60.3	55.6	46.6	39.7	35.4		
		MCDB	36.4	38.4	42.8	48.3	70.3	66.1	67.5	64.5	57.0	48.0	39.9	38.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	30.7	31.1	33.6	37.7	48.2	57.7	60.4	58.2	52.9	43.8	35.1	32.8		
		MCDB	33.3	34.1	37.2	44.5	60.2	65.4	65.9	64.0	55.0	46.2	36.8	35.3		
	5%	WB	27.6	28.3	31.5	35.5	45.3	55.1	58.8	56.5	51.4	41.4	33.2	31.1		
		MCDB	29.4	30.0	35.1	40.8	53.3	63.5	63.5	61.9	53.7	44.2	34.3	33.2		
	10%	WB	24.6	25.4	29.5	33.9	42.8	52.7	57.1	54.8	49.6	39.3	31.7	29.0		
		MCDB	26.1	27.3	32.8	39.3	50.3	61.5	62.2	60.0	52.5	42.1	32.5	30.7		
	5% DB	MDBR	17.3	19.1	21.4	18.1	14.2	14.9	15.3	16.0	13.0	11.3	14.2	16.6		
		MCDBR	15.8	15.4	19.8	19.3	24.8	21.5	20.9	22.3	16.3	12.3	10.7	13.2		
		MCWBR	14.5	13.7	15.0	10.4	9.9	8.4	7.3	8.9	9.3	7.9	9.4	12.0		
		MCWBR	14.7	15.3	17.9	16.0	19.6	18.7	14.5	17.6	11.0	10.7	9.4	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.178	0.237	0.275	0.305	0.324	0.329	0.336	0.321	0.290	0.256	N/A	N/A		
		taud	1.924	2.111	2.208	2.237	2.422	2.498	2.492	2.512	2.570	2.334	N/A	N/A		
	Ebn at Noon		37	181	243	268	276	278	272	264	246	178	N/A	0		
		Edh at Noon	3	15	24	32	30	29	28	24	18	12	N/A	0		
(44) All-Sky Solar Radiation	RadAvg		11	157	632	1248	1579	1581	1477	1062	579	192	30	0		
	RadStd		1	14	29	54	141	117	141	82	67	28	3	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page